July 13, 2026

mielski

July 14, 2026

Today, we begin our progress into answering how the differentiation operation is generalized for the quantization images,.

1 Ordering

We are familiar with the numerous ordering types by which are used in problems such as the Harmonic oscillator and the time of arrival operators.

1.1 Weyl ordering

The Weyl ordering scheme allows for the symmetric products of the operators p and q[1][2][3] for the Hamiltonian $h(p, q)$,

$$h(p, q) = q^n p^m \rightarrow \mathbf{T}_{m,n} = \frac{1}{2^n} \sum_{j=0}^{n} \binom{n}{j} q^j p^m q^{n-j} = \frac{1}{2^m} \sum_{j=0}^{m} \binom{m}{j} p^j q^n p^{m-j} \quad (1)$$

and thus for $m = 3$ and $n = 2$,

$$\mathbf{T}_{3,2} = \frac{1}{4} p^3 q^2 + \frac{1}{2} q p^3 q + \frac{1}{4} q^2 p^3 \quad (2)$$

In general, in accordance to the generalized Weyl transform[4], the quantization images could be written in the form

$$\mathbf{T}_{m,n} = \sum_{j=0}^{\min(m,n)} \frac{m! n! f^j(0)}{j!(m-j)!(n-j)!} \hbar^j p^{m-j} q^{n-j} \quad (3)$$

wherein for Weyl quantization, $f(x) = e^{ix/2}\Theta(x)$, $\Theta(x) = 1$ for the specified ordering rule. Provided this, allowing $m = 3$ and $n = 2$, the images dictate

$$\mathbf{T}_{3,2} = p^3 q^2 + 3i\hbar p^2 q - \frac{3}{2} \hbar^2 p \quad (4)$$

which are equivalent once the first definition is translated to anti-normal ordering.

---

[1] Fuuji & Suzuki. A NEW SYMMETRIC EXPRESSION OF WEYL ORDERING.

[2] Bagunu & Galapon. Quantization of the Hamilton equations of motion.

[3] Fan & Fan. WEYL ORDERING FOR ENTANGLED STATE REPRESENTATION.

[4] Domingo & Galapon. Generalized Weyl transform for operator ordering: polynomial functions in phase space